

Tech & Teach

Digital Chip Design Program

RTL Design

Task #07

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Sequential Datapath Design Part 1 - Blocking & Non-Blocking

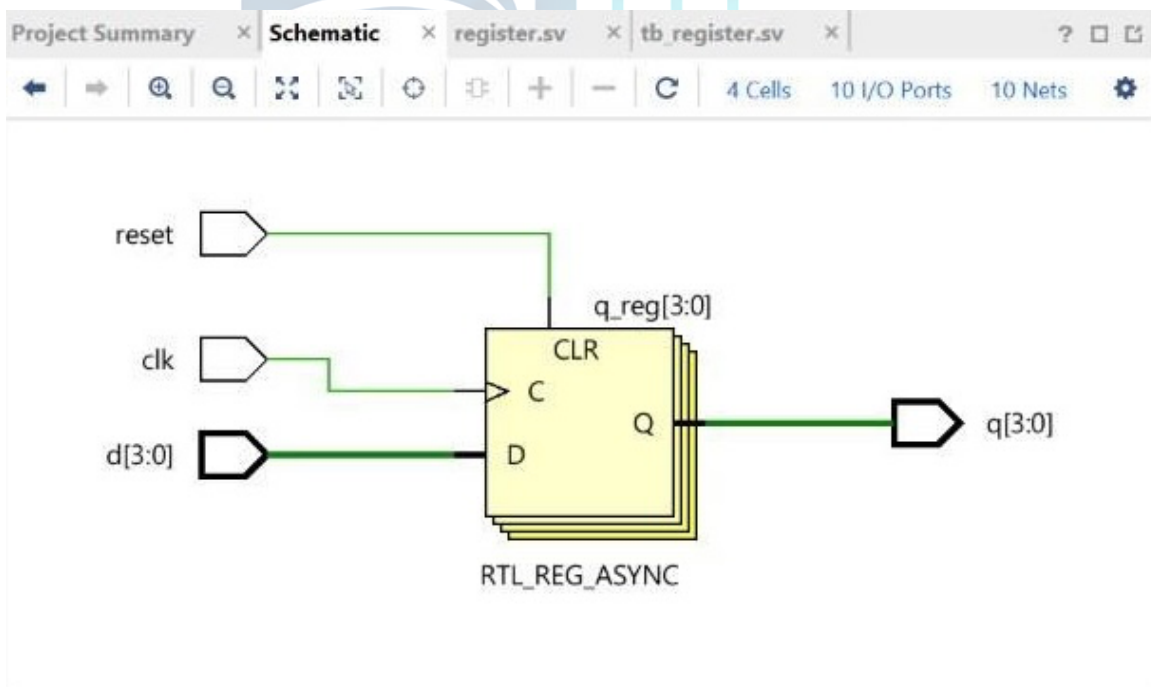
Version 1.0

Task #1.1

1.1.1.1



```
4 module register(  
5     input logic clk,  
6     input logic reset,  
7     input logic [3:0] d,  
8     output logic [3:0] q  
9 );  
10  
11 always_ff @(posedge clk or posedge reset) begin  
12     if (reset)  
13         q <= 4'b0000;  
14     else  
15         q <= d;  
16     end  
17  
18 endmodule  
19
```



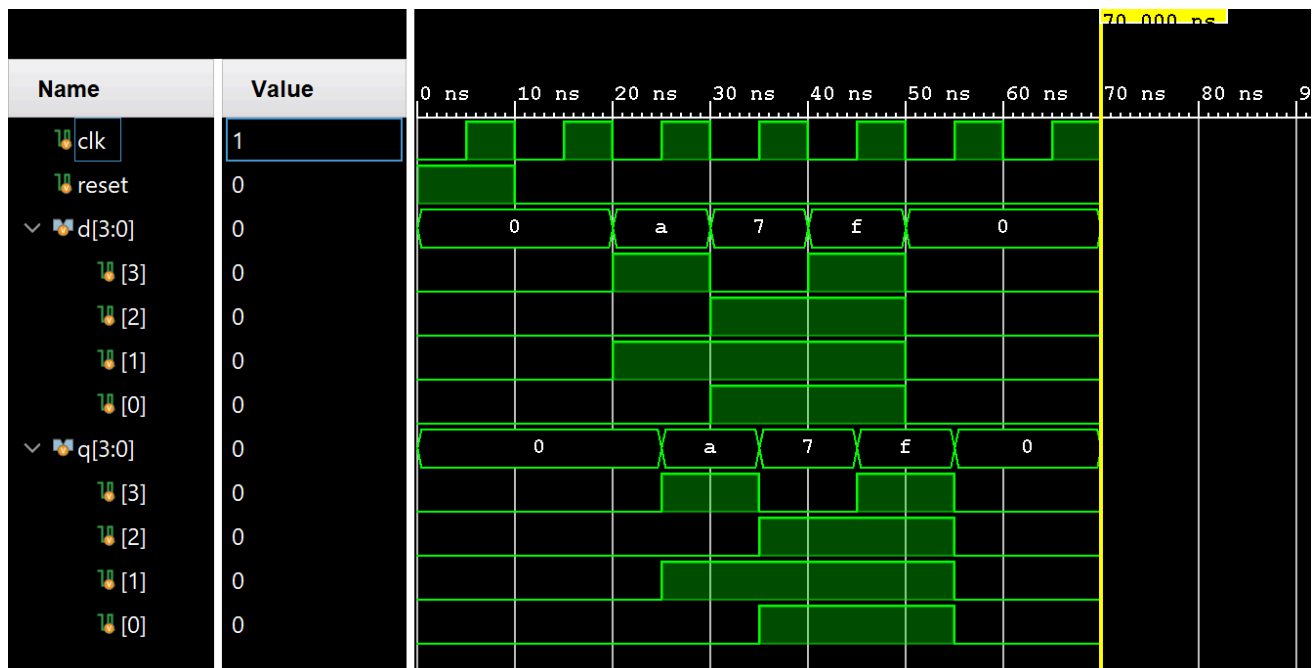
1.1.1.2

tb_register.sv *

C:/task6/register/register.srscs/sim_1/new/tb_register.sv



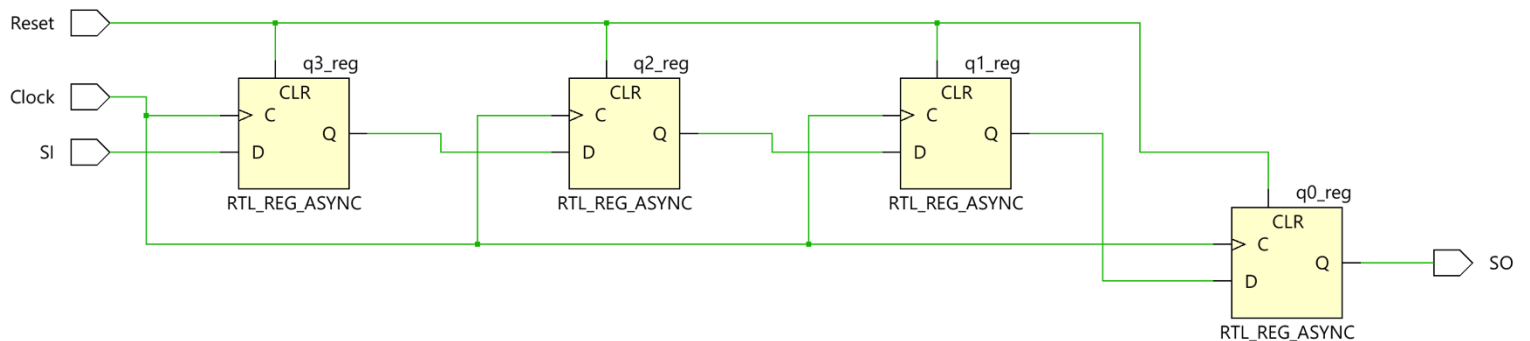
```
1 `timescale 1ns / 1ps
2
3 module tb_register;
4
5     logic clk;
6     logic reset;
7     logic [3:0] d;
8     logic [3:0] q;
9
10    register uut (
11        .clk(clk),
12        .reset(reset),
13        .d(d),
14        .q(q)
15    );
16    always #5 clk = ~clk;
17
18    initial begin
19        clk = 0;
20        reset = 1;
21        d = 4'b0000;
22
23        #10 reset = 0;
24        #10 d = 4'b1010;
25        #10 d = 4'b0111;
26        #10 d = 4'b1111;
27        #10 d = 4'b0000;
28        #20 $finish;
29    end
30
31 endmodule
```



Task #1.2

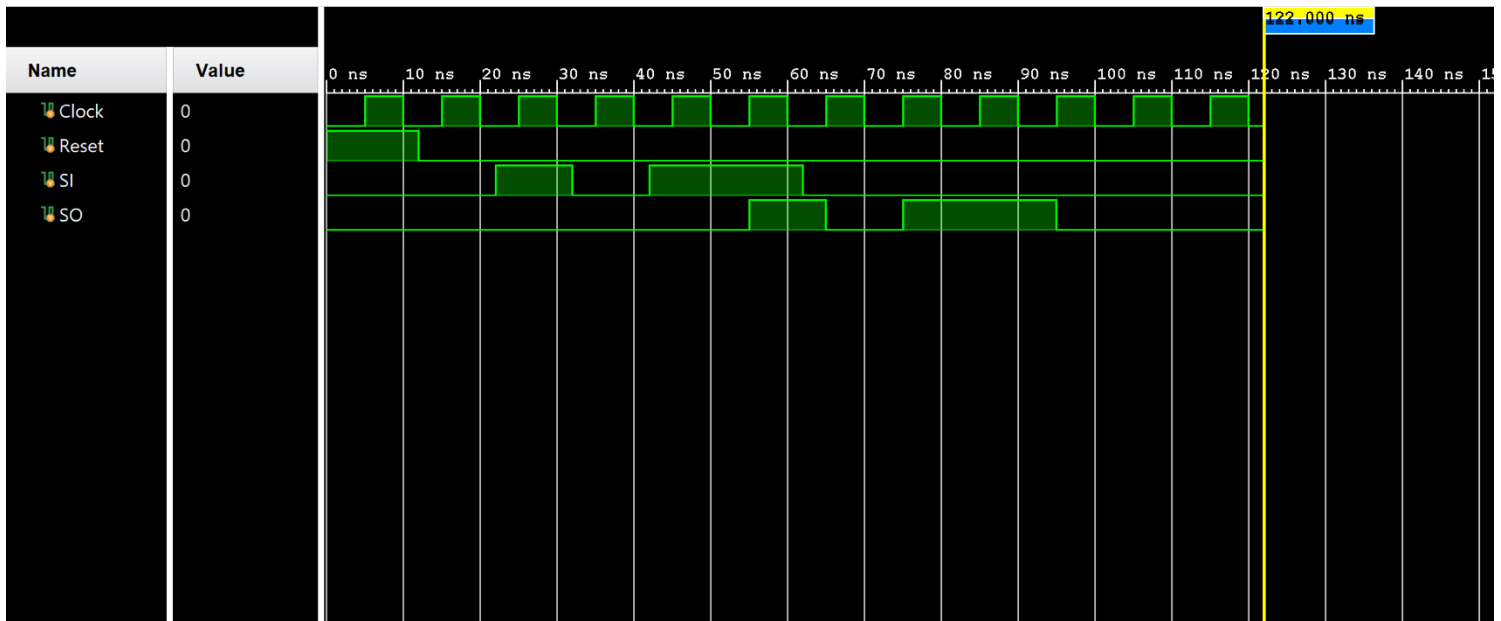
1.2.1.1

```
1 module SerialShiftReg (  
2     input  logic Clock,  
3     input  logic Reset,  
4     input  logic SI,  
5     output logic SO  
6 );  
7  
8     logic q3, q2, q1, q0;  
9  
10    always_ff @(posedge Clock or posedge Reset) begin  
11        if (Reset) begin  
12            q3 <= 1'b0;  
13            q2 <= 1'b0;  
14            q1 <= 1'b0;  
15            q0 <= 1'b0;  
16        end else begin  
17            q3 <= SI;  
18            q2 <= q3;  
19            q1 <= q2;  
20            q0 <= q1;  
21        end  
22    end  
23  
24    assign SO = q0;  
25 endmodule
```



1.2.1.2

```
1 | `timescale 1ns / 1ps
2 | module SerialShiftReg_tb;
3 |
4 |     logic Clock;
5 |     logic Reset;
6 |     logic SI;
7 |     logic SO;
8 |
9 |     SerialShiftReg dut (
10 |         .Clock(Clock),
11 |         .Reset(Reset),
12 |         .SI(SI),
13 |         .SO(SO)
14 |     );
15 |
16 |     initial begin
17 |         Clock = 0;
18 |         forever #5 Clock = ~Clock;
19 |     end
20 |
21 |     initial begin
22 |         SI = 0;
23 |         Reset = 1;
24 |         #12;
25 |         Reset = 0;
26 |
27 |         #10 SI = 1;
28 |         #10 SI = 0;
29 |         #10 SI = 1;
30 |         #10 SI = 1;
31 |
32 |         #10 SI = 0;
33 |         #50;
34 |
35 |         $finish;
36 |     end
37 | endmodule
```



Task #1.3

Project Summary × taskD7.sv × Schematic ×

C:/Users/farah/taskD7/taskD7.srscs/sources_1/new/taskD7.sv

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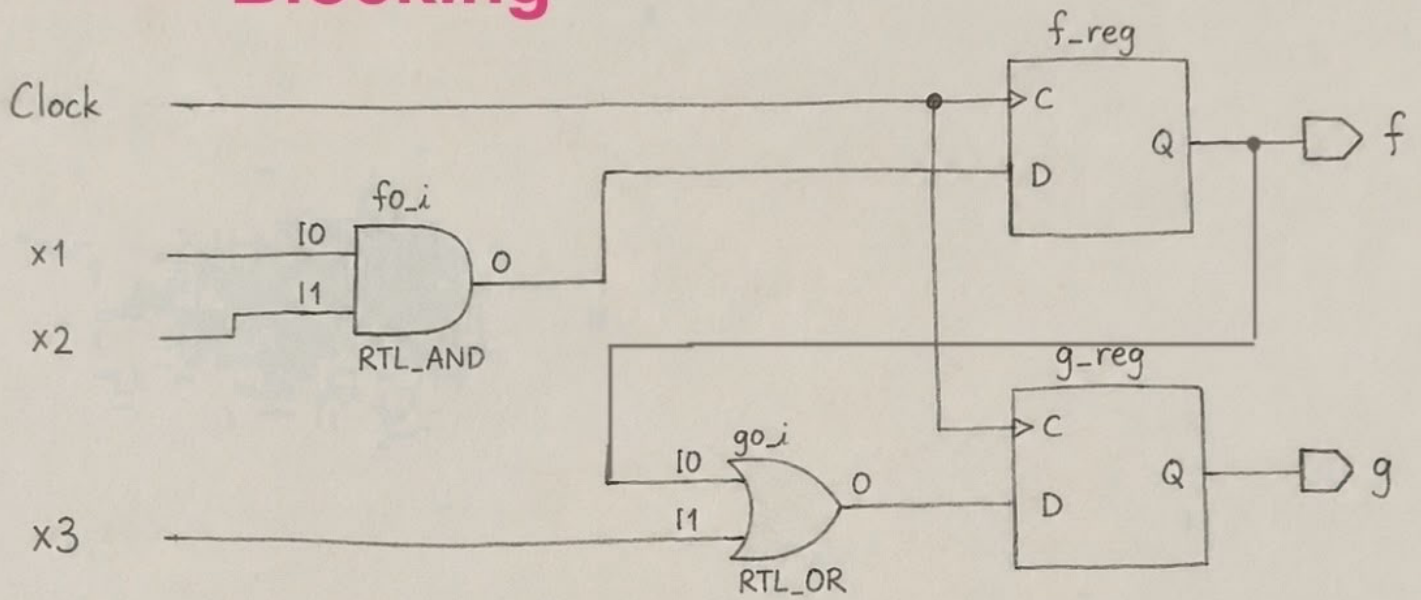
💡

```

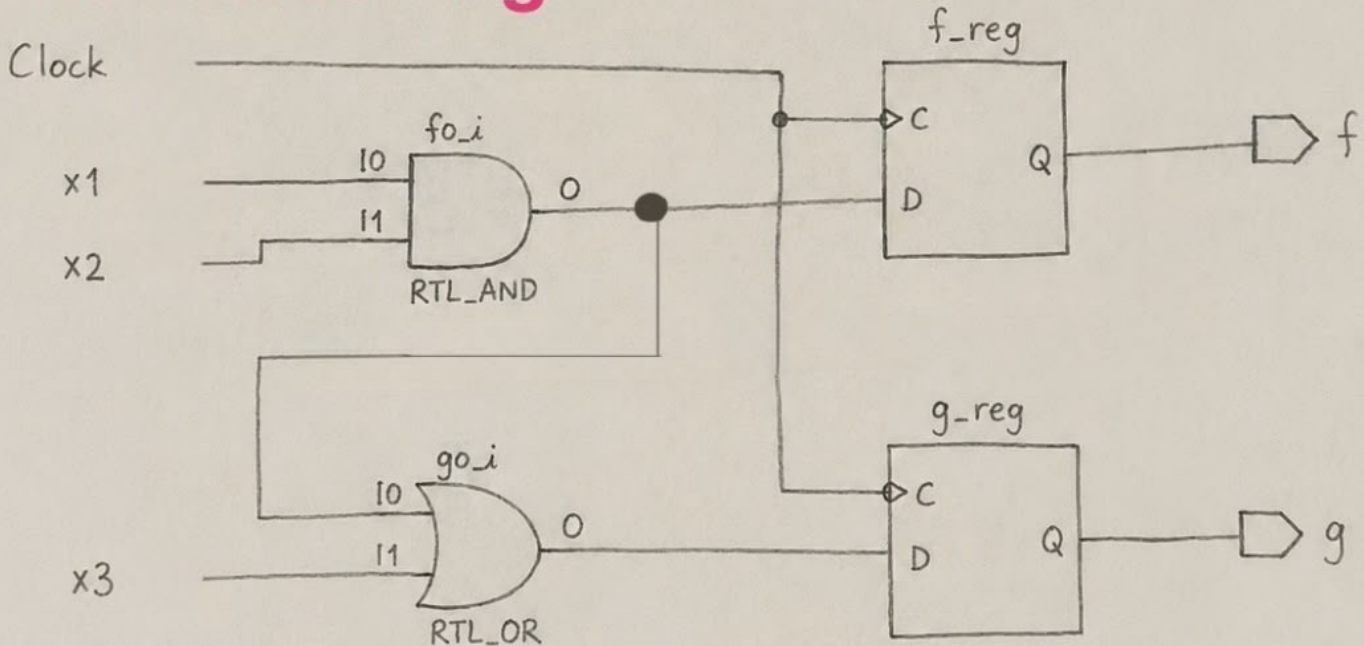
1
2 module example3 (
3     input logic x1, x2, x3, Clock,
4     output logic f, g
5 );
6     always @(posedge Clock) begin
7         f = x1 & x2;
8         g = f | x3;
9     end
10 endmodule
11
12 module example4 (
13     input logic x1, x2, x3, Clock,
14     output logic f, g
15 );
16     always @(posedge Clock) begin
17         f <= x1 & x2;
18         g <= f | x3;
19     end
20 endmodule

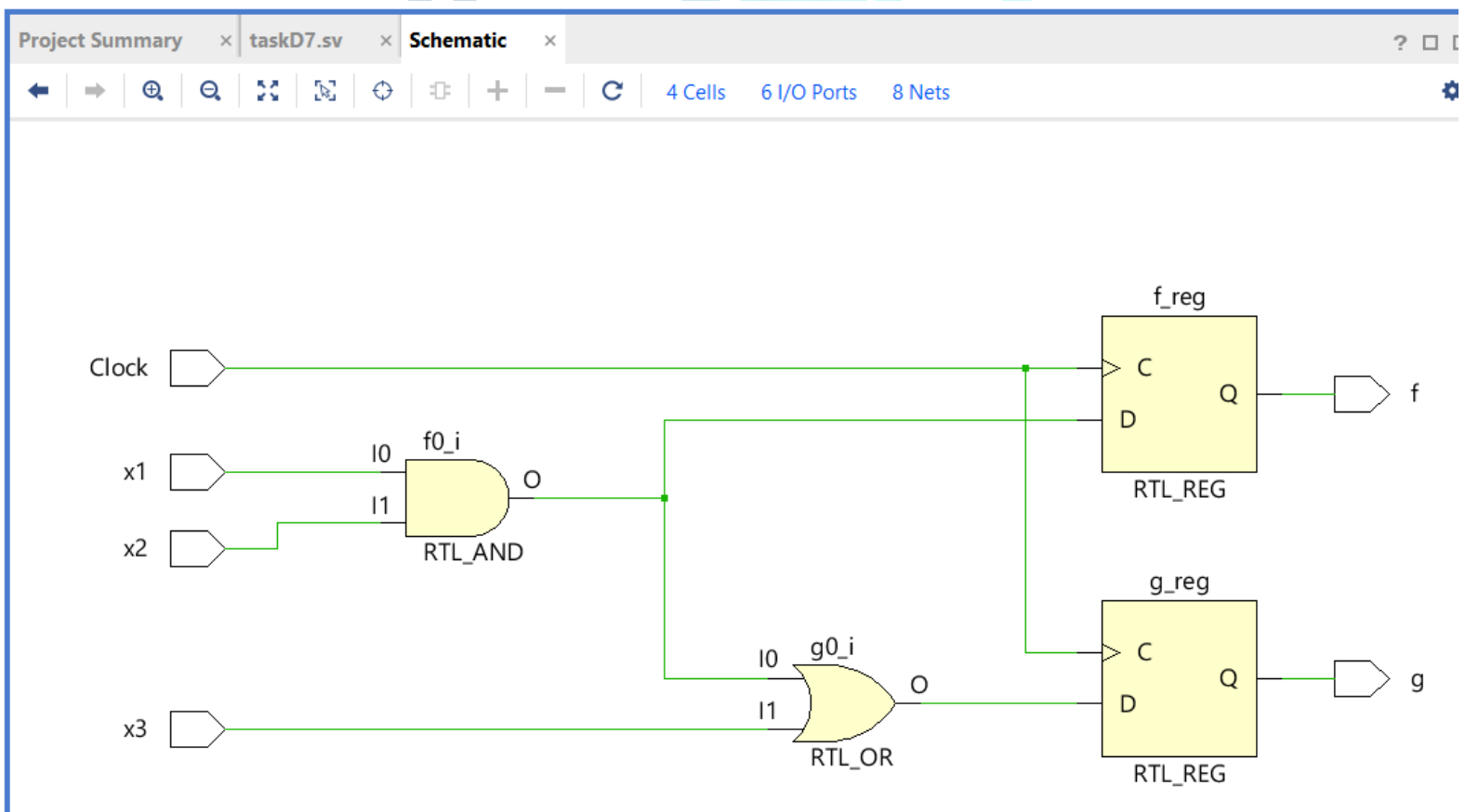
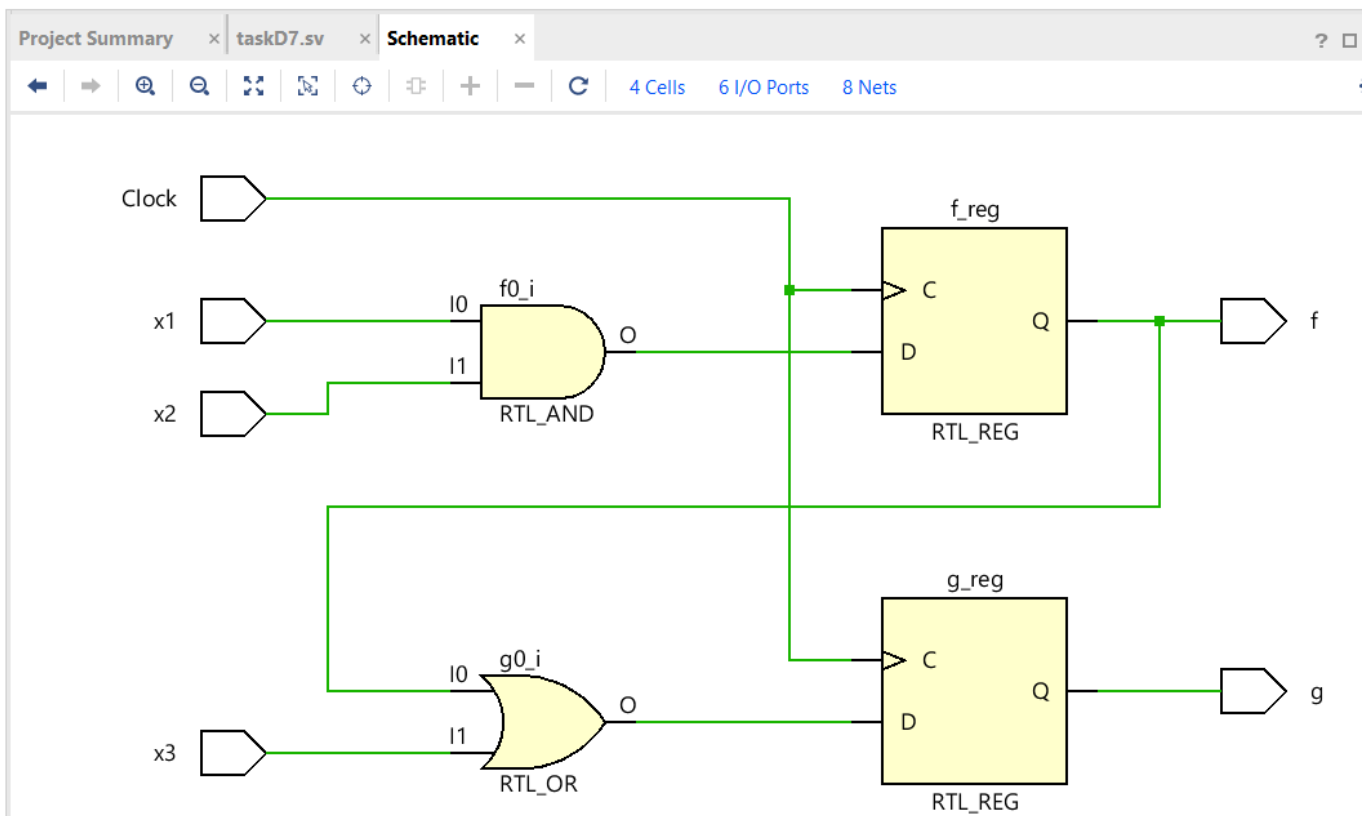
```

Blocking



Non-Blocking





Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1		✓	
Task 1.2			✓
Task 1.3	✓		